Beyond Common Factors?

The Predictive Value of Sparse Information in Euro-Area Forecasting

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Abstract

Factor models treat what the common factors leave out as noise, yet a shock to one sector may stay there and help to predict related series. We examine whether sparse forecasts of these idiosyncratic components improve factor-based predictions of euro-area headline, energy and core inflation and manufacturing production, and whether quarterly and external series, which the monthly panel lacks, strengthen them. Using EA-MD-QD and external indicators, we compare FarmPredict with factor-only, direct LASSO and autoregressive benchmarks in an expanding-window pseudo-real-time exercise respecting publication delays. Sparse corrections yield small, robust gains for core inflation with dynamic factors and manufacturing with principal components, but no systematic advantage over LASSO. Several rankings depend on the 2021–2023 crisis. Quarterly series improve manufacturing's factor forecasts; external indicators improve core factor forecasts and direct LASSO forecasts for all three inflation targets. Contrary to the intuition that sector-specific information stays in the residuals, additional information helps through the factors or direct predictors, hardly through the sparse step.

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Data and Code

Repository: `https://github.com/gianluca-tisci/factor-sparse-euro-area-forecasting`.

Data: EA-MD-QD from `https://zenodo.org/records/23064892`; external sources in Section 2.

Code: Python, with all work tracked in git.

AI: coding agents (Claude Code, Codex) helped write the code, typeset the LaTeX, organize the git, summarize the literature and edit the graphs.

Appendix: a separate file, outside the 20-page limit and not needed to follow the paper.

1 Introduction

Forecasts from large macroeconomic panels usually pass through a few common factors. A factor model summarizes what many series share in a handful of latent factors, and the forecast of any series follows from its loadings on them; what the factors leave out, the idiosyncratic component of each series, is treated as noise. Yet a shock that hits one sector or market need not be pervasive enough to load on the factors. If it is not, it stays in the idiosyncratic components of a group

of related series, which may then help to predict one another. Fan et al. (2023) build a forecast on this idea, FarmPredict: estimate the factors, forecast the common component of the target, and add a lasso forecast of its idiosyncratic component from the idiosyncratic components of all series. In their forecast of US industrial production on the FRED-MD panel (McCracken and Ng, 2016), FarmPredict beats both the lasso on the raw series and principal-component regression.

There are reasons to doubt that this carries over to other targets and panels. Luciani (2014) adds a penalized regression on the idiosyncratic components of the whole panel to a dynamic factor forecast. On US quarterly data the idiosyncratic step seldom improves the forecast, because the factors capture most of the comovement and of the dynamics; it pays off only for targets with a large and persistent idiosyncratic component. Giannone et al. (2021) find that most of the economic datasets they study do not favor sparse predictive models, and that in their simulations the lasso selects few predictors even when the true model is dense; they study the predictors in their original form, not after the factors are removed. A sectoral origin does not necessarily make a shock idiosyncratic: an energy-price shock can move many prices and quantities at once, so part of its effect may show up in the factors rather than in the residuals.

Motivation. FarmPredict can use only the information in the panel from which its factors and idiosyncratic components are estimated, and the monthly euro-area panel has gaps. It is concentrated in industry, prices and surveys; demand, employment and labor income are measured only quarterly; and it contains no commodity price, unlike FRED-MD, no measure of uncertainty and no survey of price expectations (Subsection 2.1). A mixed-frequency dynamic factor model can add both kinds of series to a monthly forecast (Subsection 3.2.1). With dynamic factors, FarmPredict can therefore use outside information in two ways: through better factors, or through the idiosyncratic components from which the sparse step selects. If shocks to one sector stay in the idiosyncratic components, as FarmPredict presumes, the outside series should help most through the sparse step. There is also an economic reason to expect a gain: our evaluation period contains the 2021–2023 surge, when euro-area energy inflation reached 10.1 percent in a month, and world energy prices are natural candidates to forecast it.

Two questions. We study FarmPredict on the euro area and ask two questions.

The first is whether combining common factors with a sparse model of the idiosyncratic components improves forecasts over models that use either one alone. It has two halves: does the sparse step improve the factor forecast, and does removing the factors first help the lasso?

The second is whether information from outside the monthly euro-area panel improves the forecasts, and whether its value depends on the model. We consider 16 quarterly national-account and labor-market series and 24 external monthly series, mostly commodity prices and price expectations, and ask whether they help through the factors, through the sparse step, or as direct predictors in a lasso without factors. The targets are headline, energy and core HICP inflation, whose idiosyncratic parts differ widely in size, and manufacturing industrial production, the euro-area counterpart of the target of Fan et al. (2023) (Subsection 2.4).

Approach. The panel is EA-MD-QD (Barigozzi et al., 2026), the euro-area counterpart of FRED-MD, and we depart from Fan et al. (2023) in three ways. First, we estimate the factors with a mixed-frequency dynamic factor model, by the EM algorithm and the Kalman smoother (Bańbura and Modugno, 2014; Barigozzi and Luciani, 2024). Quarterly series enter through the aggregation of Mariano and Murasawa (2003), and a series not yet published at the forecast date simply drops out of the likelihood, so the panel need not be cut at its last complete month. The theory of Fan et al. (2023) covers principal-component factors only, so we also estimate their version and report both versions of FarmPredict side by side: FP-PCA, for FarmPredict (PCA + VAR + LASSO), and FP-DFM, for FarmPredict (DFM + LASSO). Second, the models form a ladder in which each adds one ingredient to the one before, so that FarmPredict is compared both with its factor model alone and with LASSO, the same lasso applied to the series themselves; each model is also compared with itself as the information set grows from the monthly panel (M) to M with the quarterly series (MQ) or with the external series (ME). Third, the exercise is pseudo-real-time with revised data: at the end of each month from January 2015 to August 2026, every model is estimated again on the data published by then, each series with its own publication delay, and forecasts its target's first unpublished month, so that each forecast is a nowcast.

Findings. The answer to the first question is mostly negative. Adding the sparse step to a factor model rarely changes its forecast. The factors already explain most of the variation of headline, energy and manufacturing, at least 74% of their variance, and what they leave seldom contains a predictor strong enough for the lasso to select it. The sparse step helps in two cases only, by small amounts that hold both with and without 2021–2023. For core inflation, the target with the largest idiosyncratic component, it lowers the error of the dynamic factor model by 4.7%, as Luciani (2014) would lead one to expect. For manufacturing, it lowers the error of the principal-component model by 1.3%, and FP-PCA is the most accurate model on the monthly panel, as in the US application of Fan et al. (2023).

Nor does removing the factors first help the lasso systematically: against LASSO, the ranking depends on a few months. FarmPredict is more accurate for some targets and less for others, none of its advantages is significant, and the ranking often reverses without 2021–2023. For energy, the decisive months are November 2022 to March 2023. Energy prices then fell, while the common component still pointed up; LASSO, whose forecasts stayed close to zero, lost much less.

The answer to the second question goes against the intuition that motivates it: outside information helps, but through the factors or through LASSO, hardly through the sparse step. The quarterly series improve the dynamic factor forecast of manufacturing, by 9.5%. The external series improve the dynamic factor forecast of core inflation, by 27.1%, and LASSO for all three inflation targets, significantly only for core; for headline and energy, retailers' price expectations and the Brent oil price take the place of the target's own past among the predictors of LASSO. The sparse step of FP-DFM, in contrast, almost never selects an external series, and that of FP-PCA selects one on up to 58% of the dates, but with them its gain over the factor forecast is significant for no target. Even when the external series enter only the sparse step, with the factors of M held fixed, the forecast is never the most accurate. Apart from core inflation, these gains too come mostly from 2021–2022, and neither block makes either version of FarmPredict the most accurate model for any target.

Our results therefore qualify the case for sparse idiosyncratic forecasting on euro-area data: beyond core inflation, the factors leave little that a sparse model can use, information specific to one sector reaches the forecast through the factors or a direct lasso rather than through the residuals, and in a sample with one large shock, a few months can decide which method looks best.

Related literature. Besides Luciani (2014) and Fan et al. (2023), the closest work is FNETS (Barigozzi et al., 2024), which removes the common component of a dynamic factor model and estimates a sparse VAR for all idiosyncratic components at once, so that it models the dependence among them rather than selecting predictors for one target. Our factor step follows the EM estimation of approximate dynamic factor models by Barigozzi and Luciani (2024) and its use with missing and mixed-frequency data by Bańbura and Modugno (2014). With the same estimator, Ahn and Luciani (2026) separate common from idiosyncratic inflation in disaggregated US prices; like them, we find that the common component explains little of core inflation and more after the pandemic. On the information sets, Boivin and Ng (2006) show that adding series need not improve factor forecasts, which is why we compare information sets instead of pooling every available series.

Section 2 describes the data, Section 3 the models and the forecasting exercise, and Section 4 the results and their limitations; Section 5 concludes.

2 Data

Our main source is EA-MD-QD, the macroeconomic dataset for the euro area and its member countries of Barigozzi et al. (2026), in its September 2026 release. We use its euro-area file, 118 series of which 47 are monthly and 71 quarterly, and we add 24 external monthly series. Table 1 shows what the euro-area file contains, block by block, and what we take from it; Appendix Table A1 lists every series we use.

2.1 Choice of the Data

2.1.1 The Euro-Area Panel

EA-MD-QD. Fan et al. (2023) forecast with FRED-MD (McCracken and Ng, 2016), a large monthly panel of US series assembled for factor models. EA-MD-QD is its euro-area counterpart: Barigozzi et al. (2026) collect monthly and quarterly series, give each a transformation code that makes it stationary, adjust them for seasonality and propose a rule for outliers. The panel is therefore ready for a factor model, and we take its transformation codes and its outlier rule as our starting point instead of choosing our own, adapting the rule only where a forecast requires it (Subsection 2.3). The euro-area series start in 2000, so the whole sample lies within the monetary union. We use only the euro-area aggregates, because our targets are euro-area aggregates; the files of the ten member countries are outside our scope.

Why all monthly series and selected quarterly ones. The targets are monthly, and so are the forecasts, so the 47 monthly series form the base panel M. We use all of them, as Fan et al. (2023) use all complete FRED-MD series, so that no choice of predictors depends on the targets. The monthly block, however, is concentrated in industry, prices and surveys: 35 of its 47 series are industrial production and turnover, prices or confidence indicators (Table 1). The components of demand, employment, hours worked and labor income exist only at quarterly frequency. The quarterly series bring this information, later and less often, and a mixed-frequency factor model lets them enter a monthly forecast without inventing monthly values (Section 3); Barigozzi et al. (2026) add quarterly GDP to their monthly panel in a similar way.

Which quarterly series. We take 16 of the 71 quarterly series: GDP and its expenditure components, total employment and hours, employment in three service sectors, the wage bill and employers' social contributions, the dimensions that the monthly block lacks. The other 55 are breakdowns or ratios of series already included, credit stocks published four months after the quarter, or series that are not quarter-on-quarter growth rates of a quarterly total or average of monthly values, as the aggregation rule of the factor model requires (Mariano and Murasawa, 2003). GDP, its components, the wage bill, social contributions and hours worked are quarterly totals, and employment, following Eurostat's national-accounts definitions, is a quarterly average. For all of them the rule is an approximation, which treats the log of the quarterly value as the average of the monthly logs (Subsection 3.2.1). Since quarterly series arrive two months after the quarter ends, they can add information only on slow-moving demand and labor conditions (Subsection 4.2.1). We chose them for coverage before running the forecasts, not by searching over subsets.

2.1.2 External Series and Information Sets

Why external series. EA-MD-QD contains no commodity price, no measure of risk or uncertainty and no survey of price expectations, whereas FRED-MD includes an oil price. World

Table 1: The EA-MD-QD euro-area file and the series in our panel, by block

	EA-MD-QD		Our panel					
Block	M	Q	M	Q	E	TR3	Delay	Examples
Industrial Production and Turnover	15	0	15	0	0	2	1–3	IP and turnover by industry, car registrations
Prices	12	2	12	0	0	2	1–2	HICP headline, energy, core; PPI
Confidence Indicators	8	0	8	0	0	4	0–1	EC and OECD confidence and sentiment
Labor Market Indicators	3	18	3	7	0	2, 5	2	Unemployment; employment, hours, wages
Interest Rates	3	0	3	0	0	4	1	3- and 6-month, long-term rates
Financial Markets	3	0	3	0	0	2, 5	1–2	Exchange rates, stock prices
Monetary Aggregates	3	0	3	0	0	2	1	Currency, M1, M2
National Accounts	0	17	0	9	0	2	2	GDP, consumption, investment, trade
Labor Costs	0	8	0	0	0	–	–	Unit labor costs by sector
Credit Aggregates	0	26	0	0	0	–	–	Loans and debt securities by sector
Energy commodity prices	–	–	0	0	3	2	1–2	Brent oil, EU gas, coal
Metal prices	–	–	0	0	4	2	2	Copper, aluminum, nickel, iron ore
Food and agricultural prices	–	–	0	0	8	2	1–2	World Bank food index, grains, cocoa
Risk and supply-chain indices	–	–	0	0	3	4	1	Geopolitical risk, policy uncertainty, GSCPI
Price expectations (surveys)	–	–	0	0	3	4	0	Industry, retail and consumer surveys
HICP subindices	–	–	0	0	3	2	2	Processed and unprocessed food, NEIG
Total	47	71	47	16	24			

Notes: Number of series. M: monthly; Q: quarterly; E: external monthly. TR3: transformation code of the series we use, 2 = $100\,\Delta \log x$, 3 = $100\,\Delta^2 \log x$, 4 = x, 5 = Δx. Delay: publication delay in months, assumed for the external series. *Source:* EA-MD-QD, release 09.2026, euro-area file (Barigozzi et al., 2026); the external providers listed in Table A1.

energy and food prices are natural candidates to forecast euro-area energy and headline inflation, and our evaluation period contains the 2021–2023 surge, when energy inflation reached 10.1 percent in a month. We therefore add 24 monthly series: the prices of energy, metal, food and agricultural commodities; the geopolitical risk index of Caldara and Iacoviello (2022), the European policy uncertainty index of Baker et al. (2016) and the global supply-chain pressure index of Benigno et al. (2022); the price expectations of industry, retailers and consumers from the European Commission's surveys; and three HICP subindices (processed food, unprocessed food and non-energy industrial goods), with which the panel contains all main components of headline HICP. Each is observed in every month from 2000 to 2025 and none duplicates a series already in the panel. Before any forecast is scored, some carry more signal than the euro-area panel: in 2000–2014, with every series dated as published at the forecast date, the Brent oil price has a correlation of 0.45 with headline and 0.45 with energy inflation, against at most 0.34 and 0.28 for any monthly euro-area series. For core inflation and manufacturing the best external series is weaker than the best euro-area series (0.33 against 0.39, and 0.36 against 0.38). A correlation is not a forecast gain; Subsection 4.2 tests that for the three inflation targets.

Information sets. The four targets (headline HICP, energy HICP, core HICP excluding energy and food, and manufacturing industrial production) are members of M, so that each has a loading and an idiosyncratic component of its own, as in Fan et al. (2023). MQ adds the quarterly series to M (63 series), and ME adds the external series (71). Whether each addition improves the forecasts is the second question of the paper (Subsections 3.1.3 and 4.2).

2.2 Sample Size and Period

Estimation windows. The euro-area series start in January 2000, and the first difference costs one month, so every estimation window starts in February 2000. The external series start earlier but are cut at the same month, so that all information sets share one sample. The sample ends in August 2026, the last month of the HICP in the September 2026 release. The estimation window is expanding: it grows by one month at each forecast date, for the reasons given in Subsection 3.4. The first forecast is made in January 2015, so the first window covers about 15 years, the 2008–2009 recession and the sovereign debt crisis included; the last is made in August 2026, 140 forecast dates in all.

The size of the panel. The cross-section n is the number of series in the information set: 47 in M, 63 in MQ and 71 in ME. It follows from the choices above, and each information set changes it. The time dimension T grows with the window: from 180 monthly observations at the first forecast date to 319 at the last, and from 58 to 105 observations for the quarterly series. T is fixed by the start of the euro-area data in 2000, not by us; with so short a history we use an expanding rather than a rolling window, so that every month is kept. So n is below T in every window, but not by much: at the first date n/T is 0.39 in ME, the largest panel. Both dimensions are small next to those of Fan et al. (2023), who use 122 series and rolling windows of 480 months. Factor estimates improve as both n and T grow, but more series do not necessarily give better factors (Boivin and Ng, 2006; Barigozzi et al., 2026), which is one reason why we compare information sets instead of using every available series.

Scored months. The main comparison scores the target months up to December 2025 without those of 2020: 120

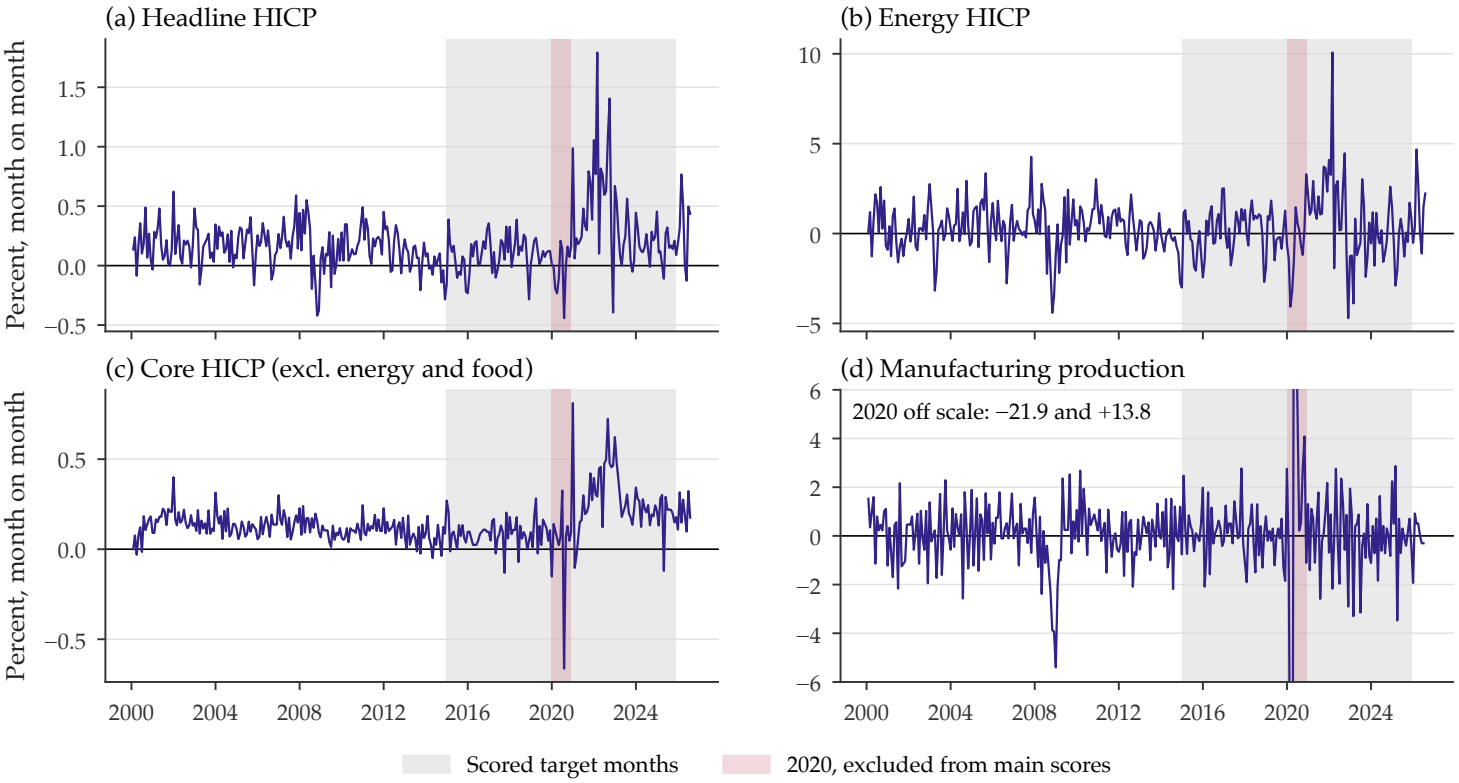


Figure 1: The four targets, month-on-month changes

Notes: Monthly change of the seasonally adjusted index, $100\,\Delta \log$, in percent. Grey: scored target months; red: 2020, not scored. Panel (d) is cut at ± 6; its two 2020 values beyond are printed in the panel. *Source:* EA-MD-QD, release 09.2026 (Barigozzi et al., 2026).

for the HICP targets and 121 for manufacturing production, whose longer delay makes its first target December 2014 (shaded in Figure 1). They cover the low inflation of 2015–2019, the 2021–2023 surge and the disinflation that followed; because a few of these months decide many results, Section 4 also scores the models without 2021–2023. We leave out 2020 from scoring, because a few of its months would dominate any comparison of squared errors: manufacturing production falls by 21.9 percent in April 2020 and rises by 13.8 percent in May 2020, against a standard deviation of 1.24 in the scored months. This concerns scoring only, whereas Ahn and Luciani (2026) drop the pandemic months from estimation: these months are exempt from the outlier rule and stay in every later estimation window, and we do not test how much they affect the forecasts. The comparison stops in December 2025, the end of the window over which the completeness of the external series was checked, so the 2026 target months enter only the robustness sample of all 140 forecasts.

Pseudo-real time. The exercise is pseudo-real-time: we use one release of revised data, the seasonal adjustment was estimated on the whole sample, and the publication delays below are fixed rules rather than historical release dates. Some external indices were also built after the fact, so their early values were not available at the time: the supply-chain pressure index was launched in 2022 (Federal Reserve Bank of New York, 2022), and the geopolitical risk index changed its search method in 2021, so its earlier values are not those published at the time. The results therefore compare forecasting rules on these retrospective information sets, not the accuracy a forecaster could have achieved at each date.

2.3 Transformations, Cleaning and Publication Delays

Transformations. We transform the EA-MD-QD series with the dataset's `TR3` codes, the benchmark of Barigozzi et al. (2026) (Table 1), the 18 external prices as monthly log changes and the 6 external indices and survey balances in levels. Each target is forecast as its month-on-month change in percent. We use the seasonally adjusted series as EA-MD-QD provides them, including the dataset authors' own adjustment of 7 of our series: energy HICP, one of our targets, and producer price indices. Of the external series, the survey balances and the HICP subindices are seasonally adjusted by their providers, whereas the commodity prices and the risk and supply-chain indices are not.

Outliers and standardization. A value more than ten interquartile ranges from the series' median is treated as missing (Barigozzi et al., 2026). Unlike the dataset paper, we compute the median and the range in each estimation window, on each series' nonzero values, and exempt 2020–2021. Nonzero values matter for iron ore and European natural gas, whose monthly change is exactly zero in 82 and 61 percent of the months of 2000–2009: on all values their interquartile range is almost zero, and the rule would flag nearly every month in which they move. Subsection 4.1.5 repeats the main

Table 2: The four targets by period

	Headline	Energy	Core	Manuf.
Mean				
2000–14	0.15	0.31	0.12	0.05
2015–19	0.10	0.08	0.09	0.07
2020	−0.02	−0.56	0.02	0.20
2021–23	0.46	1.10	0.31	0.01
2024–25	0.18	−0.08	0.21	0.00
Scored months	0.22	0.35	0.18	0.05
Standard deviation				
2000–14	0.17	1.27	0.07	1.20
2015–19	0.15	1.24	0.07	1.01
2020	0.19	1.97	0.24	9.50
2021–23	0.44	2.69	0.20	1.57
2024–25	0.13	1.32	0.09	1.28
Scored months	0.31	1.86	0.16	1.24
First-order autocorrelation				
2000–14	0.32	0.31	0.15	0.03
2015–19	0.32	0.38	−0.17	−0.18
2020	—	—	—	—
2021–23	0.24	0.26	0.40	−0.30
2024–25	0.20	0.27	−0.29	−0.54
Scored months	0.45	0.33	0.53	−0.31

Notes: Month-on-month change in percent, by period of the target month. Scored months: 2015–2025 without 2020, from December 2014 for manufacturing. *Source:* EA-MD-QD, release 09.2026, euro-area file (Barigozzi et al., 2026).

comparisons with the rule on all values. In ME, the rule flags 4 values at the last forecast date, 4 of them in currency in circulation around the 2002 euro cash changeover, and at most 7 at any date. The factor model fills the flagged values and the sparse step drops the training months that contain them; the realized values used for scoring are never changed. Each series is then centered and scaled with the moments of its estimation window.

Publication delays. Forecasts are made on the last day of each month t with the data published by then. For the monthly EA-MD-QD series we read each delay from the ragged edge of the September 2026 release: 6 series have no delay, 17 one month, 17 two and 7 three, including one month for the HICP and two for industrial production. For the quarterly series we use two months, the cautious end of the one to two months reported by Barigozzi et al. (2026). For the external series we assume zero, one and two months (3, 5 and 16 series), not verified release dates; the HICP subindices get two months although they come out with headline HICP, which leaks nothing but delays their information. Each forecast is for the target's first unpublished month, t for the HICP targets and $t-1$ for manufacturing (Subsection 3.4), and the factor model handles the ragged edge and the quarterly gaps through the Kalman filter (Section 3).

2.4 The Four Targets

Why these targets. We fixed the four targets before the forecasts and report every result. Headline HICP inflation is the measure the ECB targets, energy inflation drove the 2021–2023 surge, and core inflation is the slow, underlying component. Manufacturing production adds a real-activity target and is the euro-area counterpart of the series that Fan et al. (2023) forecast, so our results can be set beside theirs. Together the four span the cases our two questions need. On the first, the factors explain most of the variance of headline, energy and manufacturing but little of core, the case in which idiosyncratic forecasts should help most (Luciani, 2014). On the second, commodity prices are natural predictors of headline and energy inflation, but not obviously of core.

Volatility and persistence. Figure 1 plots the targets and Table 2 summarizes them by period. Energy inflation is by far the most volatile, with a standard deviation of 1.27 percentage points in 2000–2014, 2.69 in 2021–2023 and a peak in March 2022. Core inflation is smooth (standard deviation 0.16 in the scored months, against 1.86 for energy) and looks more persistent (first-order autocorrelation 0.53 against 0.33), but mostly because its level shifts, as in 2021–2023: in 2015–2019 and 2024–2025 its autocorrelation is negative. Headline inflation lies in between.

Sources and repository. EA-MD-QD, release 09.2026, is downloaded from Zenodo (`https://zenodo.org/records/23064892`). The external series come from FRED (the Brent price, the IMF primary commodity prices and the European policy uncertainty index), the World Bank Pink Sheet (food prices), the authors of the geopolitical risk index, the Federal Reserve Bank of New York (supply-chain pressure), the European Commission's business and consumer surveys (price expectations) and the ECB (HICP subindices), retrieved in November 2026. The repository stores every file as downloaded, with its source, retrieval date, hashes and terms of reuse, and its scripts write every table, figure and number in the paper.

3 Methodology

This section turns two forecasting questions into a design of paired comparisons, each built around one ingredient. Subsection 3.1 states the questions, the models and the information sets. Subsections 3.2 and 3.3 describe the estimator, and Subsection 3.4 the forecasting exercise.

3.1 Questions and Design

3.1.1 The Two Questions

The first question is whether combining common factors with a sparse model of the idiosyncratic components, the part of each series that the factors do not explain, improves forecasts over models that use either one alone. It has two halves: does the sparse step improve the factor model, and does removing the factors first help the lasso (Subsections 4.1.2 and 4.1.3)? The second is whether information from outside the monthly euro-area panel improves monthly forecasts, and whether its value depends on the forecasting method. We consider two kinds of outside information, quarterly series (Q) and external monthly series (E), described in Subsection 2.1. For each kind

we ask whether it helps through the common factors, through the idiosyncratic components, or as a predictor in a lasso without factors (Subsection 3.1.3). We answer both questions for four euro-area targets: headline, energy and core HICP inflation, and manufacturing industrial production.

Why idiosyncratic information may help. Each standardized series is the sum of two parts, $x_{it} = \chi_{it} + \xi_{it}$. The common component $\chi_{it} = \lambda_i' F_t$ is the part that moves with a few factors F_t shared by the whole panel, such as the business cycle. The idiosyncratic component ξ_{it} is what is left: the part specific to the series, or to a small group of related series. Since the factor model assumes that the two parts are uncorrelated, each can be forecast on its own, and the forecast of the series made at date t for month $t+1$ is the sum of the two:

$$\hat{x}_{i,t+1|t} = \underbrace{\hat{\chi}_{i,t+1|t}}_{\text{common part}} + \underbrace{\hat{\xi}_{i,t+1|t}}_{\text{idiosyncratic part}} . \tag{1}$$

The common part is the factor forecast: it uses the factors $F_t, F_{t-1}, \ldots$, which summarize the common movements of the whole panel. The idiosyncratic part is usually set to zero, or forecast from the target's own past only. Yet some shocks hit only a group of related series, such as one sector or one kind of price, and are too narrow to become a factor. They stay in the idiosyncratic components of that group and make them move together (Luciani, 2014). If such a shock shows up in some series before it reaches the target, the idiosyncratic components of those series carry news about the target that the factors do not contain. Forecasting the idiosyncratic part from the idiosyncratic components of all n series lets this news in; Subsections 3.2 and 3.3 estimate the two parts.

FarmPredict. With many candidates and few observations, this forecast needs a penalized regression that selects a handful of them. Fan et al. (2023) use the lasso for this step. They call a factor model combined with this sparse regression the factor-augmented regression model (FARM), and its forecast, the sum in (1), FarmPredict. In their version the factors are principal components carried forward by a VAR, and the lasso runs on the residuals of the principal-component fit. We call it FP-PCA, for FarmPredict (PCA + VAR + LASSO).

One change to the factor step. We add a second version, FP-DFM, for FarmPredict (DFM + LASSO): the same sparse step on a dynamic factor model estimated by the EM algorithm and the Kalman filter (Subsection 3.2). Principal components need every series observed in every month, which has two costs on our panel. Quarterly series cannot enter, so the principal-component models run only on the information sets M and ME. And because series are published with different delays (Subsection 2.3), the panel is cut at the last month in which every series is published, and the VAR carries the factors from there to the target month. The state-space model avoids both: quarterly series enter through their aggregation, and the latest releases of the faster series update the factors up to the target month, also in M and ME. The theory of Fan et al. (2023) is written for principal components; they note that the factor step can also be carried out with dynamic factor models, but give no theory for them. So FP-DFM is an adaptation, and FP-PCA is the version that their theory covers.

3.1.2 *A Ladder of Models*

The models. We compare models that differ by one ingredient at a time (Table 3); Figure 2a places each model by its common part and its forecast of the idiosyncratic part, and marks the contrasts listed below. All use the same targets, forecast dates and publication calendar. The benchmark is an autoregression on the target's own past (AR), which uses no panel information. We measure every gain against it because univariate forecasts are hard to beat: for US inflation, Stock and Watson (2007) find that other variables add little to a univariate forecast, and Fan et al. (2023) also report every model relative to AR. LASSO applies the lasso of the sparse step to the latest published values of the series themselves, without factors. With principal components, PCR regresses the target on the factors, the factor-only benchmark of Fan et al. (2023), and PCA-VAR forecasts the target's common component from the factor VAR and nothing else: it is FP-PCA without the sparse step. With the dynamic factor model, DFM is likewise FP-DFM without the sparse step, and DFM-AR adds to DFM an autoregression on the target's own idiosyncratic component, the standard factor forecast of that part.

Contrasts between models. Comparing FP-PCA with PCA-VAR, and FP-DFM with DFM, measures the value of the sparse step, the first half of the first question (Subsection 4.1.2). Comparing DFM-AR with DFM measures the value of the target's own persistence (Subsection 4.1.4). We do not test FP-DFM against DFM-AR: their difference, the information in other series beyond own persistence, is read descriptively, because the two models also estimate the own lags differently (least squares with a BIC order against the lasso). Comparing FP-PCA and FP-DFM with LASSO asks whether removing the common factors first helps, because de-factored series are less correlated than the raw ones (Fan et al., 2023).

3.1.3 *Information Sets*

Three information sets. We compare two additions to the monthly baseline separately. M is the monthly euro-area panel; MQ adds selected quarterly series, and ME adds external monthly series (Section 2). Each added series enters the panel from which the factors are estimated, and the candidate predictors of the sparse step and of LASSO.

Contrasts between information sets. Two contrasts follow (Figure 2b): MQ−M (quarterly data) and ME−M (external data). Each compares a model with itself, on the same target months. In FP-DFM an added series changes two things at once: it enters the panel from which the factors are estimated, which changes the common forecast and every residual, and its own residual becomes a new candidate for the lasso. A contrast of FP-DFM therefore measures the total effect of

Table 3: Models compared and the information sets on which each is estimated

Block	Model	Description	Why it is estimated	M	MQ	ME
Benchmark	AR	AR on the target	Reference; gains are measured against it	•	•	•
No factors	LASSO	Lasso on the published series	vs FarmPredict: does removing the factors help?	•	•	•
PCA factors	PCR	Target on principal components	Factor-only benchmark of Fan et al. (2023)	•	—	•
	PCA-VAR	Principal components, factor VAR	FP-PCA without the sparse step	•	—	•
	FP-PCA	PCA-VAR + lasso on all residuals	Version of Fan et al. (2023); vs PCA-VAR: sparse step	•	—	•
DFM factors	DFM	Mixed-frequency EM–Kalman factor model	FP-DFM without the sparse step	•	•	•
	DFM-AR	DFM + AR on own residual	vs DFM: own persistence	•	•	•
	FP-DFM	DFM + lasso on all residuals	Our adaptation; vs DFM: sparse step	•	•	•

Notes: •: estimated; —: not applicable, as principal components need a balanced monthly panel. M: monthly euro-area panel; Q: quarterly series; E: external monthly series. FP-PCA: FarmPredict (PCA + VAR + LASSO); FP-DFM: FarmPredict (DFM + LASSO); FarmPredict alone means both. AR uses no panel information and is estimated once per target; the fixed-factor model (Subsection 4.2.3) is estimated outside them. *Source:* Own design, after Fan et al. (2023).

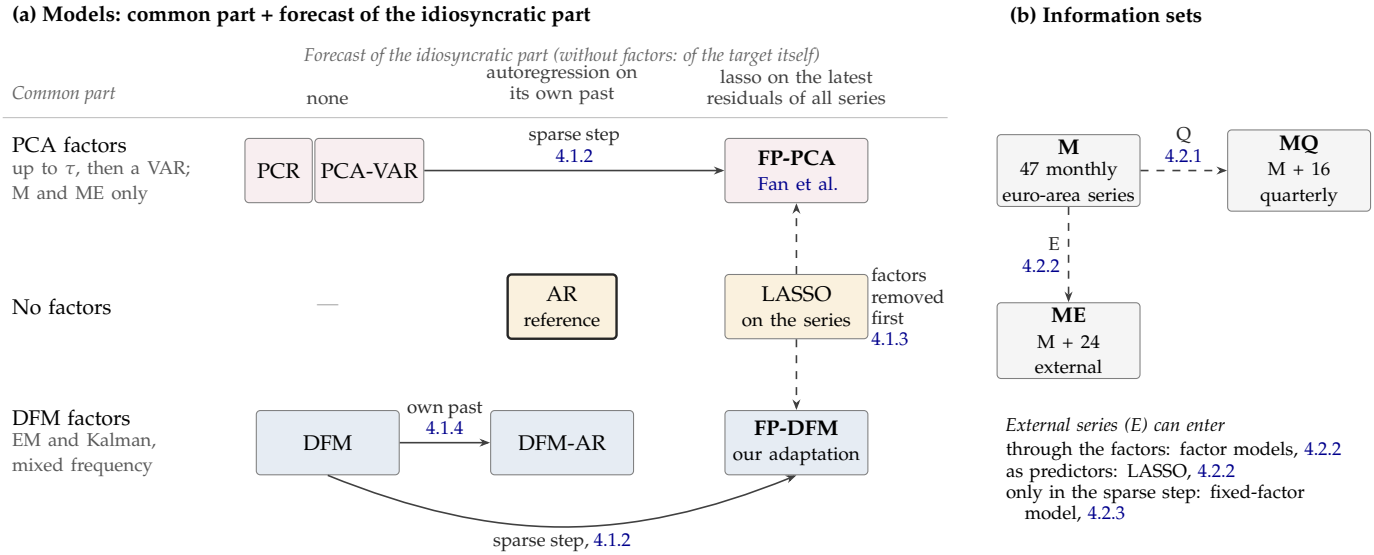


Figure 2: Models, information sets and the contrasts between them

Notes: (a) Forecast = common part (row) + forecast of the idiosyncratic part (column), as in (1); without factors, the column applies to the target itself. Arrows: contrasts, labeled with the subsection that reports them; solid: nested, one-sided Clark–West test; dashed: non-nested, Diebold–Mariano test. PCR is compared only with AR; FP-DFM against DFM-AR is read descriptively. τ: last month in which every monthly series is published. (b) Arrows: a model compared with itself as a block of series is added. Quarterly and external additions are compared separately with M. *Source:* Own design, after Fan et al. (2023).

the information and cannot say which of the two channels produced it; for the external series, the fixed-factor model described next separates them.

Three routes for the external series. An external series can help in three ways: through the factors, as a direct predictor in LASSO, or only through the idiosyncratic part, as a new candidate for the lasso. FP-DFM in ME mixes the first and the third, so we add one model that isolates the third, the fixed-factor model. It is FP-DFM in M with the same factors and common forecast, plus the external series as lasso candidates, each purged of what the factors already explain. Its gain over FP-DFM in M is, approximately, the value of the external series through the idiosyncratic part; the extra candidates also shift the lasso's training months and penalty. We run it for the three inflation targets (Subsection 4.2), since most external series are prices or price expectations (Subsection 2.1.2). We run no such experiment for the quarterly series.

3.2 Factor Model

3.2.1 *The Mixed-Frequency Model*

The state-space model. Our panel mixes frequencies and is incomplete at the end. Most series are monthly, but national accounts and employment are quarterly and observed only in the last month of each quarter, and series are published with different delays, so the latest months are incomplete. We must use all of this without inventing monthly values for the quarterly series. We therefore use a dynamic factor model in state-space form, estimated by the EM algorithm instead of the principal components of Fan et al. (2023). In it, each monthly series i is a loading on $r = 3$ common factors plus an idiosyncratic term,

$$x_{it} = \lambda_i' \boldsymbol{F}_t + \xi_{it}, \tag{2}$$

and the factors follow a joint VAR of order 2,

$$\boldsymbol{F}_t = \boldsymbol{A}_1 \boldsymbol{F}_{t-1} + \boldsymbol{A}_2 \boldsymbol{F}_{t-2} + \boldsymbol{v}_t, \tag{3}$$

where v_t is a zero-mean innovation with covariance $\mathbf{\Gamma}_v$. A quarterly series j has an unobserved monthly counterpart $\tilde{x}_{jt} = \lambda_j' F_t + \xi_{jt}$. In the last month t of each quarter it reports a weighted sum of the last five months,

$$x_{jt} = \sum_{k=0}^{4} w_k \, \tilde{x}_{j,t-k}, \qquad (w_0, \ldots, w_4) = (1, 2, 3, 2, 1). \quad (4)$$

The weights hold for the log growth of a quarterly total or average. Approximating the log of the quarterly value by the average of the logs of its three months turns quarter-on-quarter log growth into this weighted sum of monthly log growth rates, up to a factor 1/3 (Mariano and Murasawa, 2003). All selected quarterly series are quarter-on-quarter log changes of national-account and employment series, so the weights apply to each of them. External series are ordinary monthly series, so an information set only changes which series are in the panel.

Treatment of the idiosyncratic components. In the data, the idiosyncratic components ξ_{it} may be weakly correlated across series and over time. To estimate the factors we nevertheless use a simpler working model in which each is pure noise with its own variance,

$$\mathbb{E}[\xi_{it}\, \xi_{js}] = \sigma_i^2 \, \delta_{ij} \, \delta_{ts}, \quad (5)$$

where δ is the Kronecker delta. This assumption is wrong on purpose but need not spoil the factors: Barigozzi and Luciani (2024) show that the EM estimates of the loadings and the Kalman-smoothed factors remain consistent as long as the true dependence is weak. Their result is asymptotic and cannot be verified on one sample; it requires weak dependence, which holds on our panel only in part (Subsection 3.3), and it is proved for complete panels, so with mixed frequencies and missing data we borrow their estimator, not their theory. The simplification is used only to estimate the factors: the residuals $\hat{\xi}_{it}$ keep the dependence in the data, and DFM, DFM-AR and FP-DFM differ only in how they use it (Subsection 3.1.2). Since the three share the same factors, comparing them isolates the value of the idiosyncratic part.

3.2.2 *Number of Factors and Lags*

Fixed once, before the forecasts. We fix the number of factors r and the order p of the factor VAR once, on the 47 monthly series available at the first forecast date, and keep them at every date and for every target and information set. All factor models share r; the fixed p is the dynamic factor model's, while PCA-VAR chooses its own order at each date. Both are fixed before the forecasts, on the data available at the first forecast date, so no choice uses data from the evaluation period. It also keeps the factor space of the same dimension in every specification, so each contrast between models or information sets measures the ingredient it is built for rather than a change in r or p. Selecting r again at each date would instead let the dimension of the common component change between dates for reasons unrelated to these comparisons.

Table 4: Number of factors selected by each criterion at the first forecast date

Criterion	Factors	Share of orderings
Bai and Ng, IC1	15	—
Bai and Ng, IC2	15	—
Bai and Ng, IC3	15	—
Ahn and Horenstein, ER	1	—
Ahn and Horenstein, GR	1	—
Alessi et al., penalty 1 (used)	3	50%
Alessi et al., penalty 2	3	50%
Alessi et al., penalty 3	3	48%

Notes: The 47 monthly series at the first forecast date (2015-01), 177 months; at most 15 factors. ER, GR: eigenvalue and growth ratios. Penalty 1 to 3: the penalties of IC1 to IC3, scaled by the constant of Alessi et al. Share of orderings: share of 200 random orderings of the series that return the number shown. *Source:* EA-MD-QD, own calculations.

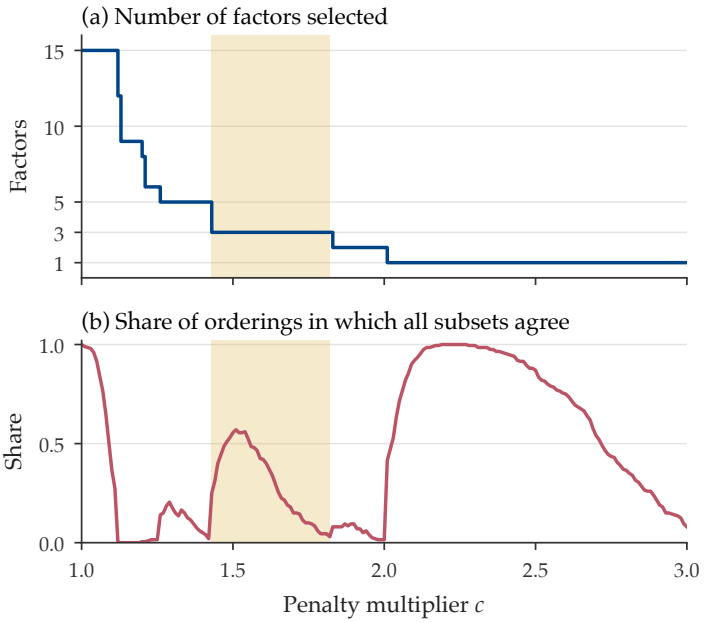


Figure 3: Number of factors selected by the criterion of Alessi, Barigozzi and Capasso as the penalty multiplier rises

Notes: The 47 monthly series at the first forecast date. Shaded: three factors. At the smallest multipliers every subset returns 15, the maximum allowed, which the criterion excludes; the short plateau at two factors is not stable across subsets. *Source:* EA-MD-QD, own calculations.

Number of factors. We choose r with the criterion of Alessi et al. (2010), designed for large macroeconomic panels. It multiplies the penalty of Bai and Ng (2002) by a constant c and selects the number of factors that stays the same over a range of c and across random subsets of the series, so that the answer does not hinge on one arbitrary penalty scale. On our panel the criterion selects 3 factors (Table 4). With all series, 3 is the answer for every c from 1.4 to 1.8 (Figure 3a), and it is the most frequent answer over 200 random orderings of the series, returned in 50 percent of them (Figure 3b); the rule to take the most frequent answer was fixed before the forecasts. The two other penalties of Bai and Ng (2002) give the same number. Barigozzi et al. (2026) select six factors on the complete EA-MD-QD panel of monthly and quarterly

series, of which our 47 monthly series are a subset.

Other criteria. The simpler criteria give no usable answer. With their fixed penalties, the information criteria of Bai and Ng (2002) select the largest number we allow, 15. The eigenvalue ratio of Ahn and Horenstein (2013) selects 1, as does the criterion of Alessi et al. (2010) at heavy penalties: both reflect how much larger the first principal component is than the others (Appendix Figure A1), and a single factor would leave the separate movements of prices, real activity and financial conditions in the idiosyncratic components.

Lags of the factor VAR. We use $p = 2$ lags in the factor VAR. At the first forecast date, the BIC of the full state-space model is lowest with 2 lags (Appendix Table A2). As a check after the fact, refitting with one, two and three lags at each December through 2025 (12 dates in all), the BIC selects two lags at 7 dates and one at the other 5, and the second lag lowers the largest root of the factor VAR from 0.98 to 0.93. Criteria computed on the estimated factors alone, which ignore the measurement equation, select more lags (Appendix Table A3).

3.2.3 *Estimation and the Common Component*

The EM algorithm. Let $\boldsymbol{\theta} = (\boldsymbol{\Lambda}, \boldsymbol{A}_1, \boldsymbol{A}_2, \boldsymbol{\Gamma}_v, \sigma_1^2, \ldots, \sigma_n^2)$ collect the parameters and $\ell(\boldsymbol{\theta})$ be the Gaussian quasi-log-likelihood of the observed data under (2)–(5). The EM algorithm maximizes it by iterating

$$\boldsymbol{\theta}^{(m+1)} = \arg\max_{\boldsymbol{\theta}} \; \mathbb{E}_{\boldsymbol{\theta}^{(m)}}\left[\ln p_{\boldsymbol{\theta}}(\boldsymbol{x}, \boldsymbol{F}) \mid \boldsymbol{x}_{\text{obs}}\right]. \tag{6}$$

The expectation (E-step) is computed by the Kalman smoother, and the maximization (M-step) has closed-form updates computed from the smoothed factors and their posterior covariance. They are least squares for the monthly series and generalized least squares for the quarterly ones. A quarterly series observes the weighted sum $\sum_k w_k \xi_{j,t-k}$ of its idiosyncratic components, and adjacent quarters share two months, so under (5) this sum has variance $19\sigma_j^2$ and covariance $4\sigma_j^2$ with the previous quarter. A missing observation (a quarterly gap, a series not yet published at the forecast date, or an outlier set to missing) drops out of the likelihood.

Re-estimation and convergence. On a complete panel EM and principal components converge at the same rates, and the EM factors are more efficient only when the idiosyncratic covariance is sparse (Barigozzi and Luciani, 2024); the practical gain of EM is in handling the gaps, the quarterly series and the publication delays (Subsection 3.1.1). At every forecast date the model is estimated again on the data published by then, starting from the principal components of the months published for every series and not from the previous date's estimates, so that each estimate depends only on the data available at that date. The iterations stop when the relative change in the log-likelihood falls below 10^{-6}, or after 3,000 iterations (other settings in Appendix A.2). Of the 420 fits used for the three information sets, 0 failed to converge and the largest root of the factor VAR is 0.97; a fit is flagged if it fails to converge, if its largest root is at least 0.98 or if it fails numerically, and 0 fits are flagged (Appendix Table A4).

Common component. For series i and a target month s, given the data published by the forecast date t, the common component is

$$\hat{\chi}_{is|t} = \hat{\boldsymbol{\lambda}}_i' \hat{\boldsymbol{F}}_{s|t}, \qquad \hat{\boldsymbol{F}}_{s|t} = \begin{cases} \text{Kalman-smoothed } \boldsymbol{F}_s, & s \le t, \\ \hat{\boldsymbol{A}}_1 \hat{\boldsymbol{F}}_{s-1|t} + \hat{\boldsymbol{A}}_2 \hat{\boldsymbol{F}}_{s-2|t}, & s > t, \end{cases} \tag{7}$$

that is, the smoothed factor if month s lies inside the estimation window and the factor VAR forecast if it lies beyond. All estimates depend on the date t of the window; we drop that index from $\hat{\boldsymbol{\lambda}}_i$ and $\hat{\boldsymbol{A}}_k$ to lighten the notation. The model is estimated on the series standardized in each window (Subsection 2.3), and the common component is then returned to each series' own units by reversing that standardization, so that from here on x_{is}, $\hat{\chi}_{is|t}$ and the residuals are in those units: the monthly change in percent for the targets. For the target, s is its first unpublished month (Subsection 3.4), the month written $t+1$ in (1). All four targets are published with a delay, so their target month lies inside the window and (7) is a nowcast from the other series. DFM, DFM-AR and FP-DFM share one fitted model per forecast date and information set. They therefore use the same $\hat{\chi}_{is|t}$ and differ only in how they forecast the idiosyncratic part.

3.2.4 *Principal-Component Models*

PCR, PCA-VAR and FP-PCA use principal components of the monthly series instead of the state-space model, with the same number of factors. Let τ be the last month in which all monthly series are published. Because principal components need a complete panel, PCA-VAR and FP-PCA estimate them on the months up to τ,

$$(\hat{\boldsymbol{\Lambda}}, \hat{\boldsymbol{F}}) = \arg\min_{\boldsymbol{\Lambda}, \boldsymbol{F}} \sum_{t \le \tau} \sum_i (x_{it} - \boldsymbol{\lambda}_i' \boldsymbol{F}_t)^2, \tag{8}$$

over the observed cells, filling internal gaps by alternating least squares. For PCA-VAR and FP-PCA, a VAR fitted to these factors carries them beyond τ, $\hat{\boldsymbol{F}}_{s|t} = \hat{\boldsymbol{b}}_0 + \sum_{\ell=1}^{\hat{p}} \hat{\boldsymbol{B}}_\ell \hat{\boldsymbol{F}}_{s-\ell|t}$ for $s > \tau$, with $\hat{p} \in \{0, \ldots, 8\}$ chosen by BIC at each date, and the common component is again $\hat{\boldsymbol{\lambda}}_i' \hat{\boldsymbol{F}}_{s|t}$. PCR instead takes the loadings from the months in which every series is observed, estimates the factors of the other months from the series observed in them, and regresses the target directly on lagged factors dated at or before τ, with the number of lags chosen by BIC. FP-PCA adds the lasso of Subsection 3.3 on the residuals of the principal-component fit. After τ, the factors of PCA-VAR and FP-PCA are therefore a VAR projection, whereas the dynamic factor model uses the latest releases of the faster series. The two blocks of factor models thus differ both in how they estimate the factors and in how they carry them to the target month, so a gap between them does not show which factor estimate is better.

3.3 Sparse Step

3.3.1 *Residuals*

We describe the sparse step for FP-DFM. At each forecast date t, the dynamic factor model fitted on the data published by then gives the residual of every series in every published month s,

$$\hat{\xi}_{js|t} = x_{js} - \hat{\chi}_{js|t}, \tag{9}$$

with the common component of (7) and the outliers of Section 2 removed. For a quarterly series the residual is the published quarterly value minus its aggregated common component (4). All residuals, including those of past months, come from the smoother of the current window, so the lasso is trained on residuals that have seen later data than the latest residual it forecasts from. This uses nothing published after t, so any harm it does shows up as worse out-of-sample forecasts and cannot make the sparse step look better than it is. FP-PCA runs the same step on the residuals of the principal-component fit (Subsection 3.2.4), for the monthly series only.

3.3.2 *The Lasso and the Forecast*

The lasso. For target i, the training response at a past date o is the residual of its first unpublished month at that date, $o - d_i + 1$. The candidate predictors $\boldsymbol{W}_{o|t}$ are the residuals of the latest four published months of every monthly series and of the latest two published quarters of every quarterly series, the target's own included, each series dated by its own publication delay. Following Fan et al. (2023), we estimate

$$(\hat{\theta}_{i0}, \hat{\boldsymbol{\theta}}_i) = \arg\min_{\theta_0, \boldsymbol{\theta}} \frac{1}{T_i} \sum_{o<t} \left(\hat{\xi}_{i,\,o-d_i+1\,|\,t} - \theta_0 - \boldsymbol{\theta}' \boldsymbol{W}_{o|t}\right)^2 + \gamma \left\|\boldsymbol{\theta}\right\|_1, \tag{10}$$

over the T_i training dates at which the response and all candidates are observed; at least 24 are required, a minimum that is never binding in our run. Candidates that are missing at date t are dropped before estimation, and an older month is never put in place of a missing one. Each candidate is standardized on the training rows, so that the penalty γ treats all candidates alike, and the intercept θ_0 is not penalized. This is the sparse regression of Fan et al. (2023), adapted to publication delays. In their balanced panel the latest residuals of every series are those of the month before the target; in ours each series enters with its latest published residuals. For a series published earlier than the target, these include the target month itself and, for manufacturing production, the month after it, so for such series the sparse step is partly a nowcast.

The penalty. The penalty γ sets how many predictors the lasso keeps: the larger it is, the fewer survive. We choose it by the modified BIC that Fan et al. (2023) use, in the form we implement,

$$\mathrm{BIC}^*(\gamma) = \ln \hat{\sigma}^2_\gamma + |\hat{S}_\gamma| \, \frac{\ln T_i}{T_i} \, \ln\ln d_W, \tag{11}$$

where $\hat{\sigma}^2_\gamma$ is the in-sample variance of the lasso's errors, $|\hat{S}_\gamma|$ the number of predictors kept and d_W the number of candidates. The first term rewards fit and the second charges a cost for each predictor, so a predictor enters only if it improves the fit enough to pay for itself (Subsection 4.1.2 shows how high this bar is). The factor $\ln\ln d_W$ raises the cost when the candidates are many, since among many candidates some fit the past well by chance. At most one predictor per 10 training dates is kept, and the coefficients are not re-estimated after selection. If no predictor is kept, the forecast is the factor forecast up to a constant.

The forecast. The forecast adds the common and the residual parts, as in (1),

$$\hat{x}_{is|t} = \hat{\chi}_{is|t} + \hat{\theta}_{i0} + \hat{\boldsymbol{\theta}}_i' \boldsymbol{W}_{t|t}, \qquad s = t - d_i + 1. \tag{12}$$

For FP-DFM, $\hat{\chi}_{is|t}$ is the common component (7) of the dynamic factor model; for FP-PCA it is the principal-component one, so the two members of the FarmPredict family differ in both parts of (12). LASSO applies (10) and (11), with the same candidate months, to the series themselves and forecasts the target directly, with no factors. DFM-AR replaces the lasso by a least-squares autoregression on the target's own residual, with up to four lags chosen by BIC.

3.3.3 *What to Expect from the Sparse Step*

What the theory covers. The guarantees of Fan et al. (2023) for this step assume principal-component factors and an exactly sparse $\boldsymbol{\theta}_i$, and they bound the estimation error of $\hat{\boldsymbol{\theta}}_i$ without giving tests on it. We therefore never call a selected predictor significant; a selection says only that the predictor survived the penalty at that date.

Are the residuals weakly correlated? The lasso works best when its candidates are not strongly correlated with one another, and removing the factors is meant to achieve this (Fan et al., 2023). On our panel it does so only in part (Figure 4a). Removing the factors lowers the average absolute correlation among the monthly series by 29 to 40 percent (at the first and the last forecast date, in M, MQ and ME), but the residuals still move in groups, led by confidence indicators, prices and industrial production and turnover. These groups may be common factors that $r = 3$ leaves out, or local factors that only the sparse step can use; Subsection 4.1.2 returns to them.

Is there anything for the lasso to find? Figure 4b shows, before any forecast is scored, how much the candidates know about each target's residual. The strongest single correlation is between 0.25 and 0.37, and between 7 and 35 of the 188 candidates lie above the noise level, against about one in twenty by chance. The lowest count, for energy inflation, is no larger than chance would produce, so with the dynamic factors the sparse step has little to work with there at this date. The lasso's bar is higher still, a correlation of about 0.20 for a single predictor (Subsection 4.1.2). A strong correlation with the idiosyncratic component also matters only as much as that component matters: it is most of core inflation but

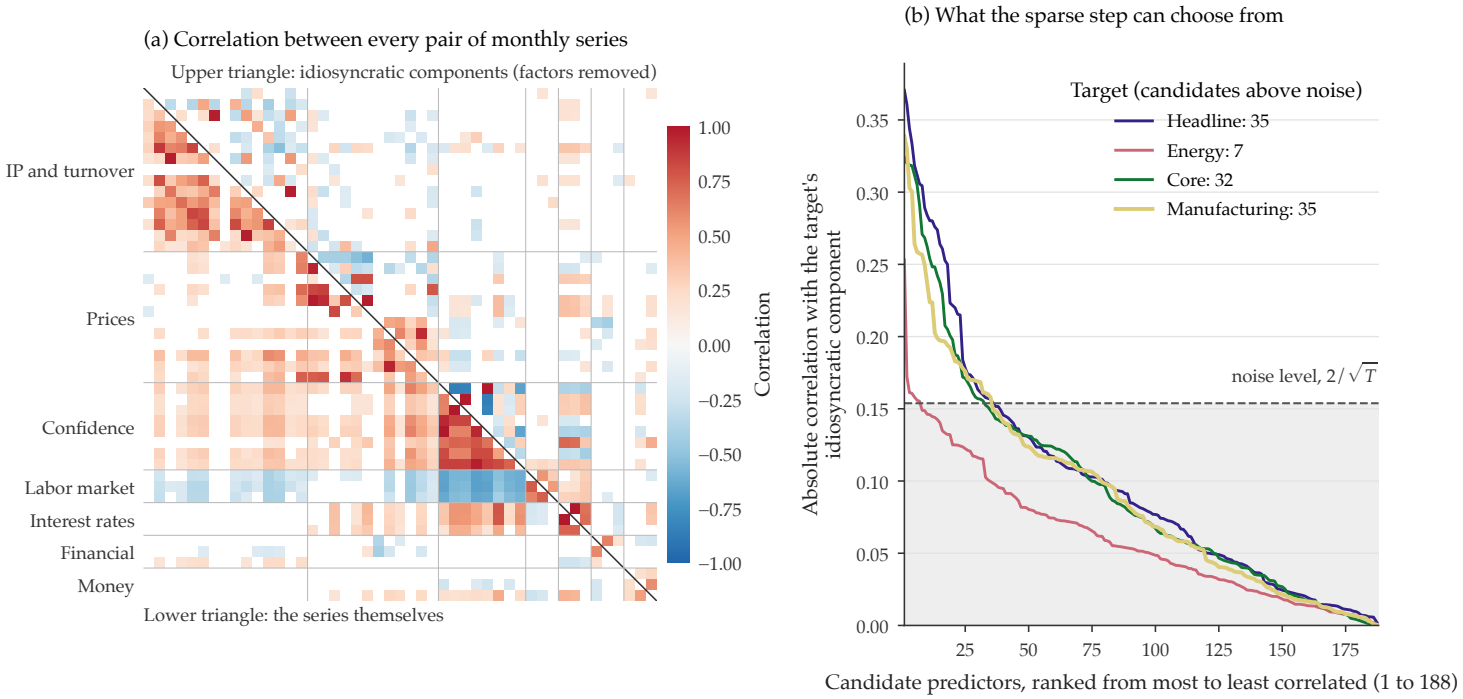


Figure 4: Correlations before and after removing the factors, and what the sparse step receives

Notes: Information set M at the first forecast date, January 2015. (a) Correlations among the 47 monthly series (lower triangle) and among their idiosyncratic components, each series minus its common component (upper triangle), by EA-MD-QD block; white: below the noise level $2/\sqrt{T}$. (b) Absolute correlation of each of the 188 candidate predictors (the idiosyncratic components of the 47 series in their latest four published months) with the target's idiosyncratic component, one at a time, sorted; a candidate unrelated to the target exceeds the noise level about one time in twenty. *Source:* EA-MD-QD, own calculations.

a small part of the other targets (next paragraph). This is a guide, not a measure: the lasso uses the candidates jointly and is re-estimated at every forecast date. Earlier evidence finds that idiosyncratic forecasts seldom help for macroeconomic series (Luciani, 2014), and our results should be read against this benchmark.

How large is the idiosyncratic part? Luciani (2014) also states when idiosyncratic forecasts do help: for targets with a large and persistent idiosyncratic component. Among our targets this is core inflation. In M, the fitted factor model assigns to the factors a median of 7 percent of the variance of core inflation at the forecast dates of 2015–2019 and 20 percent in 2021–2025, against at least 74 percent for each of the other three targets. The common share of core inflation is thus small and rises after the pandemic, as Ahn and Luciani (2026) find for US core inflation. Core inflation is therefore the target on which the sparse step has the most room, and Subsection 4.1.4 finds that it is the one where the sparse step helps the dynamic factor model.

3.4 Evaluation Design

Forecast timing. At the end of each month t from January 2015 to August 2026, we forecast every target for its first unpublished month, $s = t - d_i + 1$, where d_i is its publication delay (Subsection 2.3). This is month t for the HICP targets and month $t-1$ for manufacturing production. Each forecast is therefore a nowcast: the target month has ended, and the series published with shorter delays already cover it. It is the one-step forecast of Fan et al. (2023), dated by publication instead of by calendar month; we make no forecasts at longer horizons.

Expanding window. At each date every model is estimated again on the data published by t, in a window that starts in February 2000 and grows by one month per date. The factor-model parameters, the BIC lag orders and the lasso penalty are all chosen again from that window, so no choice uses later data; r, p and the candidate months were fixed at the first date (Subsections 3.2 and 3.3). Fan et al. (2023) use a rolling window of 480 months, but our first window holds only 180 months (Subsection 2.2): a rolling window would keep it that short and would drop the 2008–2009 recession and the sovereign debt crisis, two of the few downturns in the sample. The cost is that old and recent months weigh the same, so the models adapt slowly if the relations between the series change, for instance after the pandemic.

Scoring. Each forecast is compared with the published value of its target month, not with the cleaned value used in estimation (Subsection 2.3); the error of model a at date t is $e_t^a = x_{is} - \hat{x}^a_{is|t}$. All specifications of a target are scored on the same months, up to December 2025 without 2020: 120 for the HICP targets and 121 for manufacturing production (Subsection 2.2). The loss is the mean squared error (MSE). Since the targets differ in volatility, we compare models only within a target and report each MSE relative to AR, so that values below one beat the autoregression. The gain

of model a over model b is $100\,(1 - \mathrm{MSE}^a/\mathrm{MSE}^b)$ percent, where b is the model without the ingredient, or the same model in M for an information set. Because the MSE is an average over months, a gain can be split into the parts earned in each period, and Section 4 plots the cumulative sum of $(e_t^b)^2 - (e_t^a)^2$, which rises in the months in which a is more accurate. The model with the lowest MSE describes the sample; it is not shown to beat every other model.

Tests. For two models a and b scored on the same N dates, we test whether they differ in accuracy with

$$d_t = \left(e_t^b\right)^2 - \left(e_t^a\right)^2 + c\left(\hat{x}^b_{is|t} - \hat{x}^a_{is|t}\right)^2, \qquad S = \frac{\bar{d}}{\hat{\omega}/\sqrt{N}}, \quad (13)$$

where $\bar{d}$ is the mean of d_t and $\hat{\omega}^2$ its long-run variance; S is compared with the standard normal. When neither model contains the other, $c = 0$ and S is the two-sided test of Diebold and Mariano (1995): every model against AR, FarmPredict against LASSO, and every contrast between information sets, since added series also change the factors. When model a is model b plus a forecast of the idiosyncratic part (FP-DFM and DFM-AR against DFM, FP-PCA against PCA-VAR), the extra part adds estimation noise even when it is useless, which penalizes the larger model; $c = 1$ corrects for this (Clark and West, 2007), and the test is one-sided, with the null that the smaller model forecasts at least as well. The long-run variance uses Newey–West weights with a bandwidth of 4 months, and the months on either side of an excluded period are not treated as adjacent.

What the tests can show. The tests treat d_t as a stationary series. They do not account for the estimated factors or for the tuning repeated at every date, so their p-values are approximate. We report many contrasts and do not adjust for multiple testing, so a p-value below 0.05 is nominal evidence for that contrast alone.

Without 2021–2023. A few months of the 2021–2023 surge, when energy inflation was by far the most volatile (Subsection 2.4), can decide a comparison of squared errors. Section 4 therefore also scores the models without the target months from January 2021 to December 2023, with the same forecasts and no re-estimation; later expanding windows still contain these months. We call a gain robust when it is significant at the 5% level both on the main sample and without 2021–2023.

Robustness samples. Without re-estimating any model, we score the same forecasts on three other samples: with 2020 included, without only February 2022 to December 2023 instead of all of 2021–2023, and on all 140 forecasts, through 2026. We also re-estimate every specification with the outlier rule computed on all values instead of the nonzero ones (Subsection 2.3); Subsection 4.1.5 reports these checks. To see whether the manufacturing results extend to other industrial series, the models of M and MQ also forecast six industrial-production aggregates of the panel (Subsection 4.1.4 and Appendix Table A7).

4 Results and Discussion

Subsection 4.1 answers the first question in information set M, and Subsection 4.2 adds the quarterly and external series. Two findings stand out. First, the sparse step rarely changes the factor forecast, because the lasso often selects nothing: it helps for core inflation and for manufacturing, by small amounts that survive without 2021–2023, and most other differences between the models are decided in those three years. Second, additional information helps through different channels for different targets: quarterly series improve the factor forecasts of manufacturing, and external series improve LASSO for inflation and the factor forecasts of core inflation. Except for core, these gains too are earned mostly in 2021–2022, and neither block makes FarmPredict the most accurate model for any target. Every forecast comes from an expanding window: at each date, each model is estimated again on all the data published since 2000 (Subsection 3.4). Accuracy is the MSE relative to AR on the target months through December 2025 without 2020, so values below one beat the autoregression. AR provides a parsimonious benchmark based only on the target's own history, and univariate forecasts are hard to beat (Stock and Watson, 2007). Its competitiveness may partly reflect our one-step-ahead horizon, but we do not test this explanation against longer horizons. A step that adds a forecast of the idiosyncratic part to a factor model is tested with the one-sided Clark–West statistic, and a change of information set, or FarmPredict against LASSO, with the Diebold–Mariano statistic (Subsection 3.4); the tests are not adjusted for multiple comparisons.

4.1 Factors and Sparsity

4.1.1 The Forecasting Ladder

We address the first question (whether combining factors and sparsity improves forecasts over using either alone) using the monthly panel M, where we can compare both versions of FarmPredict in a setting close to that of Fan et al. (2023). Subsection 4.2 then examines how forecast performance changes when quarterly and external series are added to the information set. Table 5 groups the models by how they estimate the factors, making it easier to compare factor-only forecasts with their extensions. PCR regresses the target on the principal components, FP-PCA and FP-DFM add the sparse step to PCA-VAR and DFM, DFM-AR adds an autoregression on the target's own idiosyncratic component instead, and LASSO applies the same lasso to the published series themselves. The first question therefore has two halves: does the sparse step improve the factor model (Subsection 4.1.2), and does removing the factors help the lasso (Subsection 4.1.3)? Because 2021–2023 is an exceptional period (Subsection 2.2), the right half of the table repeats the comparison without those three years.[1] We call a gain robust when it is significant at the 5% level both on the full sample and without them (Subsection 3.4).

[1] The exclusion applies only to forecast evaluation. Models estimated after 2023 still use the 2021–2023 observations in their expanding training samples.

Table 5: The forecasting ladder in M: MSE relative to AR

		Full sample				Without 2021–2023			
Block	Model	Headline	Energy	Core	Manuf.	Headline	Energy	Core	Manuf.
Benchmark	AR	1.000	**1.000**	1.000	1.000	1.000	**1.000**	1.000	1.000
No factors	LASSO	1.004	1.056	1.313	0.947	**0.974**	1.174	1.043	**0.833**
PCA factors	PCR	0.975	1.083	**0.930**	0.917	1.194	1.226	1.142	0.943
	PCA-VAR	0.968	1.189	1.179	0.912	1.277	1.326	1.068	0.968
	FP-PCA	0.942	1.070	1.176	**0.900**	1.280	1.358	1.060	0.949
DFM factors	DFM	0.893	1.248	1.078	1.547	1.140	1.037	1.138	1.521
	DFM-AR	**0.886**	1.184	0.933	1.592	1.088	1.022	**0.995**	1.569
	FP-DFM	0.897	1.255	1.026	1.544	1.138	1.037	1.074	1.519

Notes: MSE relative to AR, information set M; lowest per target and sample in bold. Full sample: target months to December 2025, without 2020; without 2021–2023: also without January 2021 to December 2023 (for manufacturing, a descriptive split). Tests: Appendix Tables A5 and A6. *Source:* Own calculations.

Which model is best. No single model performs best across all four targets (Table 5). On the full evaluation sample, DFM-AR is best for headline inflation, while PCR and DFM-AR are almost tied for core inflation. FP-PCA is best for manufacturing, and no model outperforms AR for energy inflation.

The rankings change when 2021–2023 is excluded from evaluation. LASSO becomes the best model for both headline inflation and manufacturing. For headline, none of the factor models beats AR; for core, only DFM-AR remains approximately as accurate as AR. These changes show that several models owe their full-sample advantage to their performance during 2021–2023.

4.1.2 Does the Sparse Step Improve the Factor Model?

Adding the sparse step to the factor model changes little (Figure 5, panels a and b). With the dynamic factors it barely moves the forecast of headline, energy and manufacturing in any information set, even where the change is significant. Its one robust gain with the dynamic factors is for core inflation: 4.7% in M and 5.5% in MQ, where the quarterly series enter through our mixed-frequency model (Subsection 3.2.1); in ME, with the external series, it disappears, for reasons that Subsection 4.2 explains. With principal components the lasso selects something more often for headline and energy (Table 6) and the sparse step moves the forecast more, but its large gain for energy comes entirely from 2021–2023; only its small gain for manufacturing, 1.3%, is robust.

Why the sparse step adds little. Within information set M, the sparse step adds little because the factors already explain most of the targets' variance and the lasso sets a high bar for the remaining predictors. First, in M, the fitted DFM assigns at least 74% of the variance of headline, energy and manufacturing to the factors, compared with 7% for core inflation in 2015–2019 and 20% in 2021–2025 (median shares across forecast dates; Subsection 3.3). This leaves a smaller share of variance for the idiosyncratic model for the other three targets. Second, a residual predictor enters the lasso only if it lowers the in-sample error variance by about 3.9%, the penalty of the modified BIC (11) on our samples, which for a single predictor means a correlation of about 0.20 with the target's residual. What is left after the factors is weak and moves mostly within groups of related series, such as prices, surveys or interest rates (Subsection 3.3.3), so few predictors clear the bar: once the strong common cycle of the euro-area series is removed, what remains seldom carries news about the target. The lasso of FP-DFM selects nothing on 46.7% of the forecast dates for core, the lowest share, and on 92.5% for energy (Table 6); on those dates FP-DFM adds only the intercept of the lasso, an average of past residuals, to its factor forecast.

4.1.3 Does FarmPredict Outperform LASSO?

In M, neither approach consistently outperforms the other (Table 5). On the full evaluation sample, FP-DFM has a lower MSE than LASSO for headline and core inflation, but a higher MSE for energy and manufacturing. FP-PCA outperforms LASSO for every target except energy. None of these advantages of FarmPredict is significant at the 5% level. Without 2021–2023, the FP-DFM rankings reverse for all three inflation targets, and LASSO outperforms FP-PCA for every target, significantly so for headline and energy.

The models also select different predictors (Table 6). LASSO often uses the target's own past, producer prices and leading indicators of activity, whereas FarmPredict selects a small set of predictors for the variation left after removing the factors. This difference does not translate into a consistent forecasting advantage.

Much of the comparison is driven by 2021–2023 (Figure 6). LASSO's losses in 2021–2022 favor FP-DFM for headline and core. For energy, the opposite occurs: between November 2022 and March 2023, FP-DFM predicts increases while prices fall, whereas LASSO predicts changes close to zero (Figure 7). LASSO therefore makes smaller errors without anticipating the decline; outside those five months, FP-DFM is more accurate.

4.1.4 Core Inflation and Manufacturing

Core inflation and manufacturing illustrate how bridging factor and sparse models can improve forecasts over a factor-only model. In M, the sparse step delivers robust gains over DFM for core and over PCA-VAR for manufacturing. These

Table 6: What each lasso selects in M

	LASSO		FP-DFM		FP-PCA	
Target	Predictor	%	Predictor	%	Predictor	%
Headline	*No predictor*	14	*No predictor*	50	*No predictor*	34
	Own past	84	3-month interest rate	31	Consumer confidence	33
	PPI, consumer goods	84	Construction sentiment	12	Long-term interest rate	31
	PPI, durable consumer goods	36	Consumer confidence	8	6-month interest rate	17
Energy	*No predictor*	18	*No predictor*	92	*No predictor*	32
	Own past	68	Business confidence	6	Consumer confidence	33
	Stock prices	42	PPI, energy	6	PPI, energy	32
	PPI, energy	32	HICP, industrial goods	2	6-month interest rate	27
Core	*No predictor*	3	*No predictor*	47	*No predictor*	77
	3-month interest rate	63	Long-term interest rate	39	Economic sentiment	15
	PPI, consumer goods	34	Construction sentiment	24	Long-term interest rate	8
	Long-term interest rate	19	3-month interest rate	15	—	
Manufacturing	*No predictor*	45	*No predictor*	65	*No predictor*	55
	Industrial confidence	55	Own past	34	3-month interest rate	29
	Car registrations	50	PPI, energy	16	Production, capital goods	18
	Stock prices	50	Youth unemployment	2	Core HICP	9

Notes: Information set M. For each lasso: share of forecast dates on which it selects no predictor, and its three most frequent predictors with their selection shares. LASSO selects among the published series, FP-DFM and FP-PCA among idiosyncratic components; own past: the target's latest value (LASSO) or latest idiosyncratic component. Names shortened; full names in Table A1. Target months to December 2025, without 2020. *Source:* Own calculations.

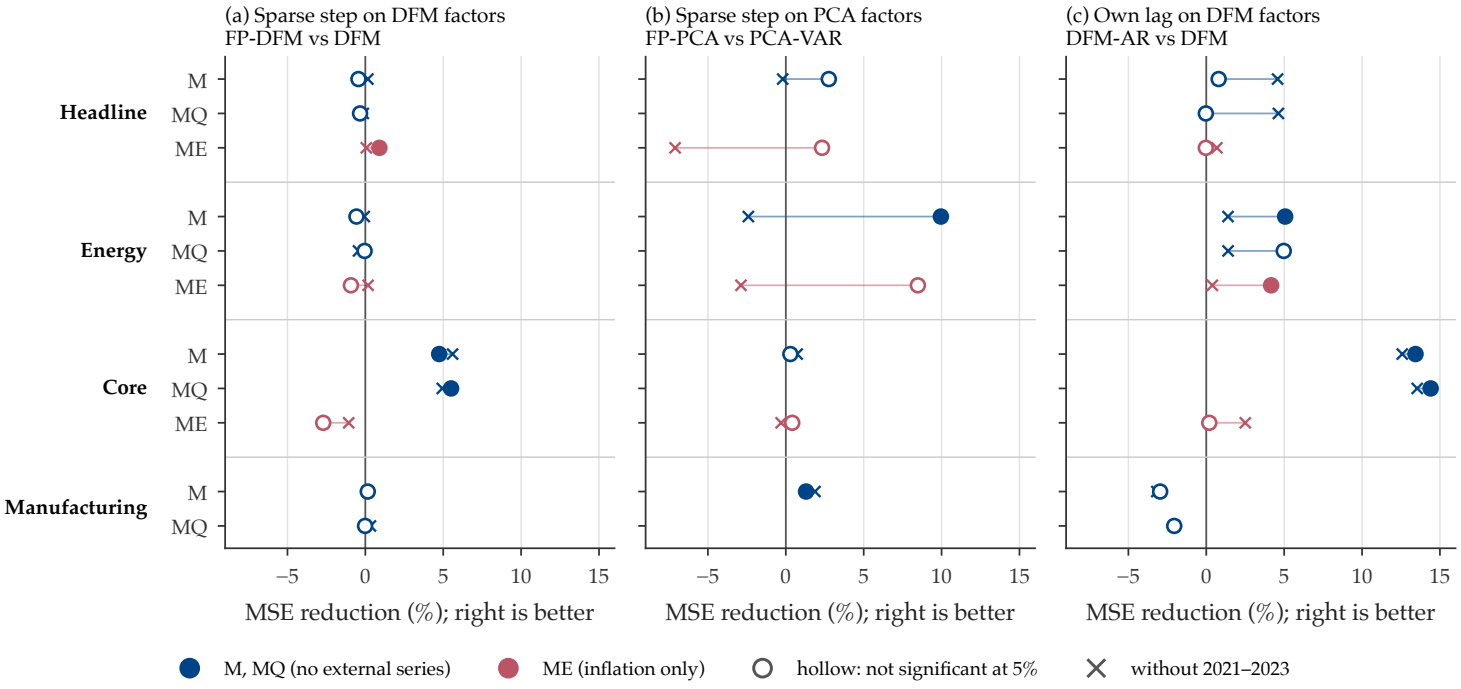


Figure 5: The gain from each ingredient, by information set

Notes: MSE reduction from adding the ingredient, within each information set, so values to the right of zero are gains. Dots for the full sample, hollow if the one-sided Clark–West $p \geq 0.05$; crosses without 2021–2023, whose tests for M are in Appendix Table A6. PCA-VAR and FP-PCA are not estimated with quarterly series, so panel b has no MQ results. Details in Appendix Table A5. *Source:* Saved paper experiment.

cases help clarify what idiosyncratic information adds, without implying that FarmPredict consistently outperforms the other models.

Two channels for core inflation. FP-DFM selects mainly interest rates and confidence indicators, particularly the long-term interest rate and construction sentiment, but never core's own lags (Table 6). These predictors may contain information about financial conditions and demand that is not fully captured by the common factors. This is a possible interpretation of their selection, rather than evidence of a causal mechanism.

DFM-AR captures a different source of information by explicitly including the target's own idiosyncratic lags. It lowers the DFM's MSE by 13.4%, with gains concentrated from 2023 onward as core inflation declines (Figure 6); this improvement

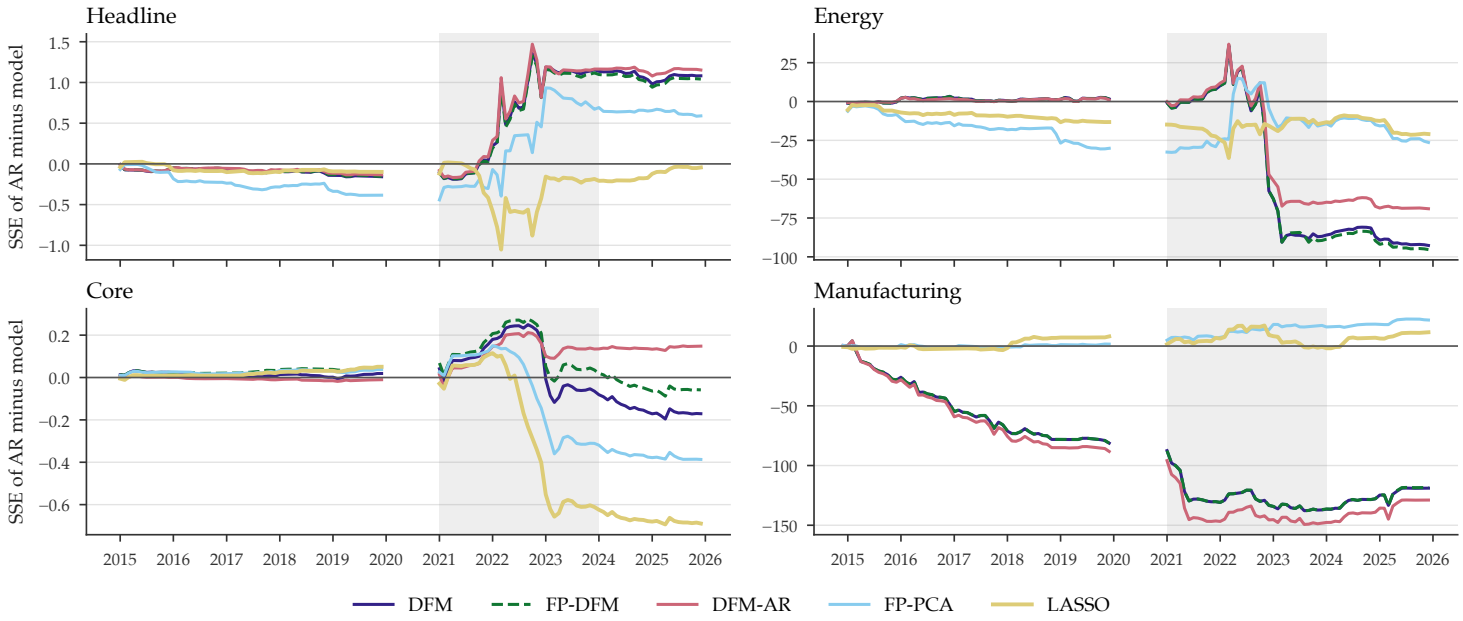


Figure 6: When each model gains on AR

Notes: Cumulative sum of the squared error of AR minus that of each model, information set M, in squared percentage points; the line rises when the model beats AR. Scales differ across panels; DFM and FP-DFM almost coincide. 2020 not scored; 2021–2023 shaded. *Source:* Own calculations.

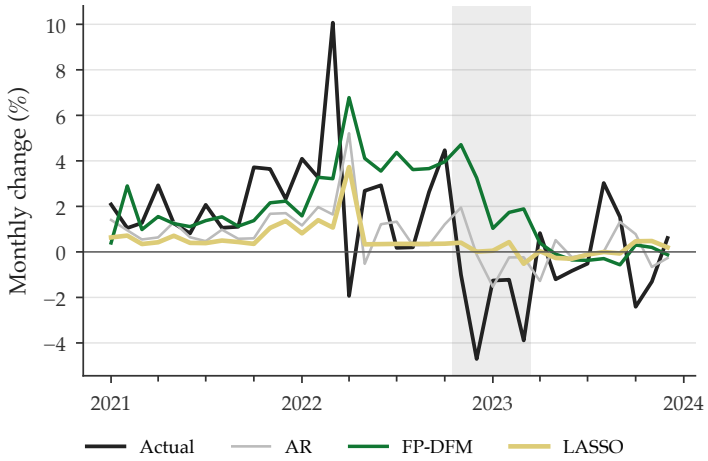


Figure 7: Energy inflation and its forecasts, 2021–2023

Notes: Energy HICP, monthly change in percent, and its forecasts by target month, information set M; FP-DFM almost coincides with DFM. Shaded: November 2022 to March 2023. *Source:* EA-MD-QD and own calculations.

is robust too. Thus, both information from other series and own-component dynamics can improve the factor-only forecast. Nevertheless, PCR and DFM-AR remain more accurate than FP-DFM on the full evaluation sample.

Manufacturing and the original application. Manufacturing provides a comparison with the US industrial-production application of Fan et al. (2023). FP-PCA is the most accurate model in M on the full evaluation sample, with an MSE 10.01% below AR, compared with the 9.20% reported in their application, where FarmPredict's advantage over the lasso is also small and is not tested. Its sparse step selects mainly short-term interest rates and capital-goods production (Table 6).

The improvement over PCA-VAR is robust, but the advantage over LASSO is not statistically significant and is earned in 2021–2023: without those years, LASSO performs best. Among the additional industrial aggregates, FarmPredict is the most accurate model only for capital goods (Appendix Table A7). DFM, meanwhile, performs substantially worse than AR from the start of the evaluation sample. Since DFM and PCA-VAR differ in both factor estimation and dynamics, this gap cannot be attributed to the factor estimates alone.

4.1.5 *Robustness*

We check whether the gains from adding the sparse step to DFM for core inflation and to PCA-VAR for manufacturing survive alternative evaluation samples and a different outlier rule (Subsection 3.4). For core, the sparse step lowers the error of the dynamic factor model by 4.0% when 2020 is included, by 2.6% when only February 2022 to December 2023 is left out, and by 4.7% when the outlier rule is computed on all values rather than the nonzero ones (Subsection 2.3), always significantly; the own-lag correction holds in the same samples. For manufacturing, the sparse step on principal components holds with the narrower exclusion and the other outlier rule; with 2020 included, its MSE reduction remains positive and increases to 32.9%, but is not statistically significant, because a few pandemic months dominate the comparison. Appendix Table A8 lists the most accurate specification of each target on all 140 forecasts and with the narrower exclusion.

4.1.6 *Answer to the First Question*

Combining factors and sparsity yields selective gains, rather than a general improvement. In M, the sparse step robustly improves DFM for core inflation and PCA-VAR for manu-

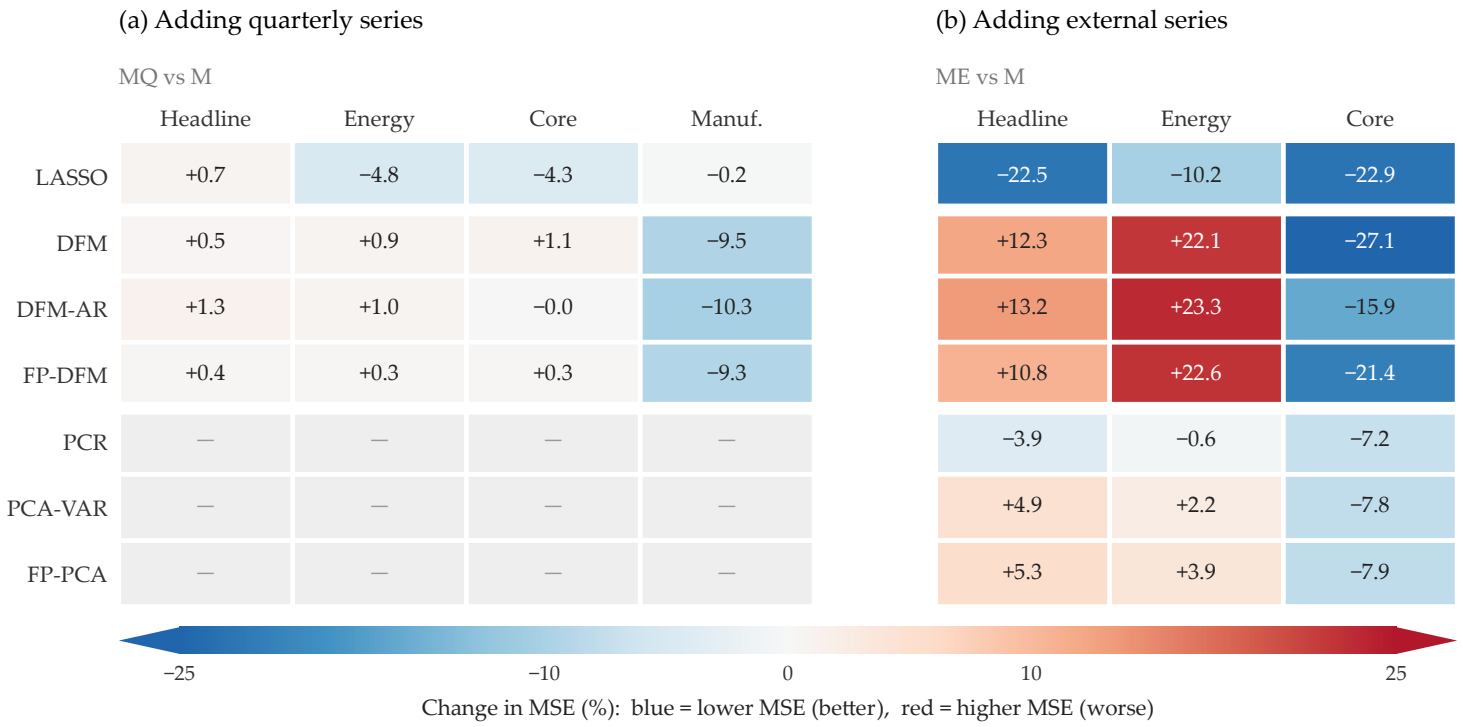


Figure 8: Changes in forecast MSE when quarterly or external information is added

Notes: Change in the MSE of each model against the same model in M; target months through December 2025, excluding 2020. Panel (a) includes all four targets; panel (b) focuses on the three inflation targets. Negative percentage changes indicate improvement; positive changes indicate deterioration. —: PCR, PCA-VAR and FP-PCA are not estimated with quarterly series. These are comparisons of forecasting procedures, not causal information effects. *Source:* Saved paper experiment.

facturing. The core result is consistent with the relevance of idiosyncratic information discussed by Luciani (2014). However, these gains over factor-only models do not translate into a systematic advantage over LASSO: none of FarmPredict's advantages is statistically significant at the 5% level, and several rankings depend on 2021–2023.

4.2 Quarterly and External Information

We consider Q for all four targets, then focus on E for the three inflation targets, restricting the external-series comparison to M and ME along the three routes of Subsection 3.1.3. Figure 8 shows how the MSE of each model changes when the quarterly (panel a) or the external series (panel b) are added, Figure 9 shows when these changes are earned, and Table 7 compares the ways of using E: in the factors, only in the sparse step, or in LASSO. Each model is compared with itself in M by the two-sided Diebold–Mariano test. Figure 5 compares the gains from the sparse step and the own-lag correction within each information set.

4.2.1 *Quarterly Series: MQ versus M*

For the factor models, the question is whether more balanced coverage of macroeconomic activity improves their forecast of the common component. Adding Q reduces manufacturing DFM MSE by 9.47% (Diebold–Mariano $p = 0.03$), with a similar reduction for FP-DFM ($p = 0.03$). Most of this gain, 6.3 of its 9.5 percentage points, is earned in 2021–2022 (Figure 9). A plausible interpretation is that quarterly national accounts and labour-market indicators broaden the representation of real activity relevant to manufacturing, with the demand and labor dimensions that the monthly panel lacks (Subsection 2.1.1). This gain does not coincide with a larger common variance share: across the same forecast dates, the median model-implied share falls from 86.8% in M to 85.5% in MQ in 2015–2019, and from 93.1% to 89.9% in 2021–2025, with three factors throughout. The variance decomposition and out-of-sample accuracy therefore measure different things: quarterly information improves the forecast without increasing the share of manufacturing variation assigned to the common component. The improvement does not close the gap with the strongest monthly benchmark: FP-PCA in M remains the best manufacturing forecast across the M and MQ specifications. Inflation factor forecasts change little in economic magnitude and do not improve; LASSO gains for energy and core, but not significantly ($p = 0.28$ and 0.28).

4.2.2 *External Series: ME versus M*

E includes commodity prices, supply-chain and risk indices, expectations and additional HICP components (Subsection 2.1.2). It improves core inflation factor forecasts (DFM: $p = 0.02$), and DFM-AR in ME becomes the most accurate specification for core. Headline and energy factor forecasts instead worsen, though not significantly (DFM: $p = 0.57$ and 0.34), whereas LASSO improves for all three inflation targets ($p = 0.08$, 0.09 and 0.03). This is consistent with Boivin and Ng (2006) and Bai and Ng (2008): enlarging a predictor panel need not yield factors more useful for a specific outcome

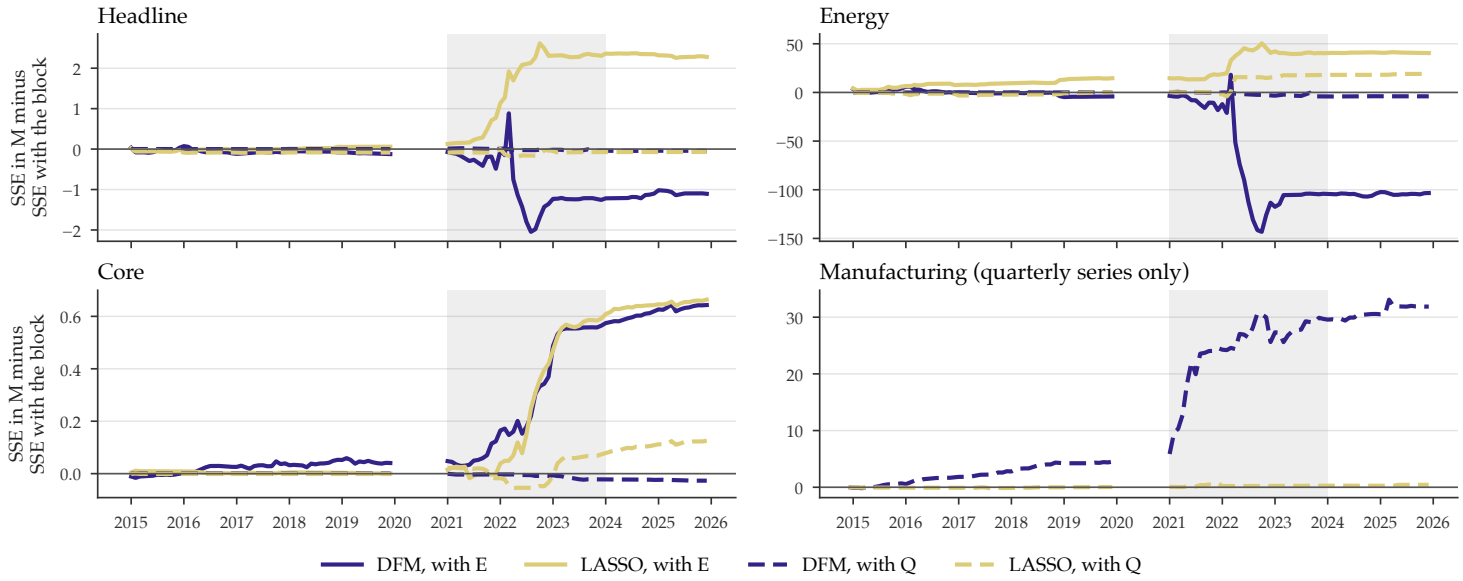


Figure 9: When the quarterly and external series help

Notes: Cumulative sum of the squared error of each model in M minus that of the same model with the block, in squared percentage points; a rising line means that the block lowers the error in those months, and the scales differ across panels. E (solid lines) is studied for the three inflation targets and Q (dashed lines) for all four. 2020 is not scored; 2021–2023 shaded. *Source:* Saved paper experiment.

Table 7: Where the external series enter: MSE relative to AR

	Full sample				Without 2021–2023			
Target	Without E	E in factors	E in sparse step	E in LASSO	Without E	E in factors	E in sparse step	E in LASSO
Headline	0.897	0.993	0.896	**0.778**	1.138	1.123	1.114	**0.954**
Energy	1.255	1.538	1.231	**0.948**	**1.037**	1.064	1.040	1.047
Core	1.026	**0.807**	0.880	1.012	1.074	0.960	0.966	**0.928**

Notes: MSE relative to AR; lowest of the four in each row and sample in bold. The first three columns are FP-DFM: in M without E; in ME with E entering factor estimation and the sparse step; and with the M factors and common forecast fixed and external residuals added only to the sparse step. E in LASSO is direct LASSO in ME. Full sample: matched target months through December 2025, excluding 2020; the second sample additionally excludes January 2021–December 2023. Source: saved paper and fixed-factor experiments..

(Subsection 2.2). For headline and energy most of these changes happen in 2021–2022 (Figure 9): LASSO earns 22.2 of its 22.5 percentage points of gain for headline in those two years, and the factor forecasts lose more in them than over the whole sample. Core is the exception: its gains keep growing after 2022, and only 13.9 of the DFM's 27.1 points come from 2021–2022.

What LASSO selects. In LASSO, the external series replace the target's own lags for headline and energy: these enter on 101 and 81 of 120 dates in M, respectively, but never in ME. Retail selling-price expectations are selected at every ME headline origin, and lagged Brent price changes at every ME energy origin; Brent is the external series most correlated with both targets before the evaluation period (Subsection 2.1.2). Reconstructing the saved LASSO forecasts confirms that the external terms contribute to the predictions, rather than merely being selected; removing their contributions while holding coefficients fixed raises MSE for all three targets. By contrast, the sparse step of FP-DFM picks an external residual on at most 7% of the ME dates for any target; that of FP-PCA picks one more often, on up to 58% of the dates for headline, but in ME its gain over PCA-VAR is significant at the 5% level for no target (Appendix Table A5), so in this monthly comparison the external series act mainly through the factors or in LASSO, rather than through the sparse step.

Core inflation through the factors. For core FP-DFM, the external gain comes from the factors, not from the sparse step. From M to ME, DFM MSE falls by 27.09%, compared with 21.40% for FP-DFM. The fitted model also assigns more of core's variance to the common component: across the same forecast dates, the explained share rises from 6.6% in M to 15.4% in ME in 2015–2019, and from 19.7% to 33.9% in 2021–2025, keeping three factors throughout. This is a within-model variance decomposition, not a measure of out-of-sample forecast accuracy; the reduction in forecast MSE provides separate evidence of predictive improvement. The sparse step selects no predictor, and so adds only its intercept, on 96.7% of ME dates, compared with 46.7% in M. Thus the improved common forecast accounts for the full model's net gain, while the incremental sparse correction partly offsets it.

4.2.3 *External Series Only in the Sparse Step*

The fixed-factor models keep, at each forecast date, the factors and the common forecast of FP-DFM in M and add the external series only to the sparse step, as the residuals of a regression of each external series on those factors and an intercept (Subsection 3.1.3). For headline and energy, preserving the original common forecast avoids much of the deterioration observed when E also enters factor estimation: the fixed-factor models substantially outperform the fully expanded ME FP-DFM models (Table 7). This supports the interpretation that the expanded factor representation is less useful for these targets, although the fixed-factor model also changes the candidate predictors, the training rows and the penalty, so it does not isolate the effect of the factors. In the primary sample, however, the recovery is insufficient to outperform LASSO with external predictors.

Four ways to use the external series. Table 7 sets the four ways of using the external series side by side. On the full sample LASSO is the most accurate for headline and energy and the factors are for core, while adding E only to the sparse step is never the most accurate, and for headline it is only as accurate as FP-DFM without E. Without 2021–2023 LASSO is the most accurate for core as well.

Energy without 2021–2023. The energy ranking changes when the 2021–2023 crisis period is excluded from evaluation. Without those years, the fixed-factor model reduces MSE relative to LASSO in ME by 0.64%, a small difference that we do not test. Thus the primary-sample advantage of LASSO over the fixed-factor model depends on the crisis period. Nevertheless, the fixed-factor model remains worse than AR and DFM-AR in M (their strong performance may reflect the short, one-step-ahead forecast horizon). External residuals are selected, particularly retail price expectations in the fixed-factor energy models, but their incremental forecasting benefit is modest. Outside the crisis, the fixed-factor model's advantage over LASSO rests on the retained common forecast: the sparse correction does not improve that base DFM forecast.

4.2.4 *Answer to the Second Question*

Additional information helps, but which block helps depends on the model. The quarterly series improve only the dynamic factor forecasts of manufacturing, and the external series improve LASSO for all three inflation targets and the factor forecasts of core inflation. In both cases the information works through the factors or in LASSO, hardly through the sparse step. As in Subsection 4.1, most of these gains are earned in 2021–2022; only those for core keep growing afterwards (Figure 9). Neither block makes either version of FarmPredict the most accurate model for any target.

4.3 Limitations

Our conclusions are subject to several limitations. First, the exercise is pseudo-real-time only: it uses revised data, full-sample seasonal adjustment and assumed publication delays rather than historical vintages. Second, the evaluation sample is short and dominated by one episode: with 120 scored months for the HICP targets and 121 for manufacturing, small differences in accuracy are hard to detect, the tests are not adjusted for multiple comparisons, and several rankings depend on 2021–2023. The pandemic months, although not scored, remain in every later estimation window, and we do not test how much they affect the factors. Third, the limited value of the idiosyncratic components is conditional on our specification: the modified-BIC penalty of the lasso, the number of factors ($r = 3$) and the one-month horizon; our results therefore do not establish that idiosyncratic components are generally uninformative. Finally, Fan et al. (2023) note that the factor step can also be carried out with dynamic factor models but give no theory for them, so their guarantees do not cover our mixed-frequency EM implementation; and neither predictor selection nor the fixed-factor comparisons identify causal mechanisms.

5 Conclusion

Factors and sparsity. The first question was whether combining common factors with a sparse model of the idiosyncratic components improves on either alone. Adding the sparse step to a factor model seldom helps (Figure 5): the factors account for at least 74% of the variance of headline, energy and manufacturing, and what remains rarely contains a predictor strong enough to enter the lasso. It helps for core inflation, the target with the largest idiosyncratic component, lowering the MSE of the dynamic factor model by 4.7%, as Luciani (2014) would lead one to expect, and for manufacturing, where FP-PCA improves on AR by 10.01%, close to the 9.20% of Fan et al. (2023). Removing the factors before the lasso yields no systematic gain: neither version of FarmPredict is significantly more accurate than LASSO at the 5% level, and the ranking often reverses without 2021–2023; in Fan et al. (2023), too, the advantage over the lasso is small and not tested.

Outside information. The second question was whether information from outside the monthly panel improves the forecasts, and through which route. It helps through the factors, for manufacturing with the quarterly series (9.5%) and for core with the external series (27.1%), and through the direct lasso for all three inflation targets, significantly only for core, but not through the sparse step (Figure 8). FP-DFM almost never selects an external series, FP-PCA selects one on up to 58% of the dates without a significant gain, and adding the external series to the sparse step alone never gives the most accurate forecast (Table 7). This runs against the intuition that motivated the question. Sector-specific information is not necessarily idiosyncratic predictive information: a shock that originates in one sector may load on the factors, or be strong enough for a lasso on the target itself, as the Brent price and retailers' price expectations are for energy and headline inflation.

Relation to the literature. Two findings for US data carry over to mixed-frequency factors and an enlarged euro-area panel: idiosyncratic forecasts rarely help (Luciani, 2014), and more series need not yield factors better suited to a given target (Boivin and Ng, 2006; Bai and Ng, 2008). They do not show whether the idiosyncratic components carry no information or dense, weak information that the lasso fails to detect (Giannone et al., 2021).

Implications. A sparse idiosyncratic step is therefore a target-specific tool rather than a default extension of a factor model; outside information should be tested in the factors and as direct predictors, not only in the residuals; and in samples with one large shock, rankings should be reported with and without it. Subsection 4.3 discusses the limitations of the exercise.

Extensions. A dense or Bayesian model of the idiosyncratic components, such as the spike-and-slab regression of Giannone et al. (2021), would tell these two possibilities apart. Since different models work better for different inflation components, combining forecasts of energy, food and core may also improve on forecasting headline inflation directly. Finally, other factor dimensions and penalties, real-time vintages and longer horizons would show how far our conclusions extend.

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